Linear Algebra with Applications in Automatic Control

School of Automation,
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Fall Semester

Review

- Section 2.3 Ideals of polynomials
  - On Polynomial Ideals, Their Complexity, and Applications (1995)
  - Used in this section to define the g.c.d, and l.c.m, which is used to define coprime factors
- Section 2.4 Factorization of polynomials
  - Coprime factorization of polynomials into the product of powers of a series of prime
  - Prime is of degree one, the field is algebraically closed, C is closed, R is not
- Section 2.5 Matrix polynomials
  - Polynomials with coefficients in the ring of matrices set

Outline

- Section 2.6 Unimodular matrix polynomials and the Smith canonical form

- Section 2.7 Elementary divisors and equivalence of matrix polynomials

- Section 2.8 Ideals of matrix polynomials and coprimeness

2.6 Unimodular matrix polynomials

Z. Song

- Definition of rank

Definition 2.11 A λ-matrix $A(\lambda)$ is of rank r, written rank $A(\lambda) = r$, if it has a nonzero minor of order r ($r \geq 1$) with all minors of order $r+1$ (if any) being zero. A zero matrix is of rank zero.

- In linear algebra, a **minor** of a matrix ***A*** is the determinant of some smaller square matrix, cut down from ***A*** by removing one or more of its rows or columns. Minors obtained by removing just one row and one column from square matrices (**first minors**) are required for calculating matrix cofactors, which in turn are useful for computing both the determinant and inverse of square matrices.

2.6 Unimodular matrix polynomials

Z. Song

- Definition of unimodular matrix polynomials

Definition 2.12 $A(\lambda) \in \mathcal{F}^{n\times n}[\lambda]$ is said to be invertible if there exists a λ-matrix $B(\lambda) \in \mathcal{F}^{n\times n}[\lambda]$ such that

$$A(\lambda)B(\lambda) = B(\lambda)A(\lambda) = E_n \qquad (1)$$

$B(\lambda)$ is called the inverse of $A(\lambda)$ and denoted $B(\lambda) = A^{-1}(\lambda)$. In other words, a λ-matrix is invertible if (1) holds for a λ-matrix $B(\lambda)$.

Theorem 2.20 $A(\lambda) \in \mathcal{F}^{n\times n}[\lambda]$ is invertible if and only if $\det A(\lambda)$ is a nonzero scalar in $\mathcal{F}$.

Defintion 2.13 An invertible λ-matrix is called a unimodular matrix.

2.6 Unimodular matrix polynomials

Z. Song

- Definition of elementary operations

Definition 2.14 The following operations on a λ-matrix are called elementary row (column) operations.

1. Interchange the ith row (column) and the jth row (column).

2. Multiply the ith row (column) by a nonzero scalar $\gamma \in \mathcal{F}$.

3. Add to the jth row (column) the ith row (column) multiplied by an arbitrary polynomial $\varphi(\lambda) \in \mathcal{F}[\lambda]$.

2.6 Unimodular matrix polynomials

Z. Song

- Elementary matrices and the properties

1. Interchange the ith row (column) and the jth row (column); $P(r_i \leftrightarrow r_j)$, $(P(c_i \leftrightarrow c_j))$
2. Multiply the ith row (column) by a nonzero scalar $\gamma \in \mathcal{F}$; $P(\gamma r_i)$ $(P(\gamma c_i))$
3. Add to the jth row (column) the ith row (column) multiplied by an arbitrary polynomial $\varphi(\lambda) \in \mathcal{F}[\lambda]$ $P((\varphi)r_i + r_j)$ $(P((\varphi)c_i + c_j))$

- Invertible

$$P^{-1}(r_i \leftrightarrow r_j) = P(r_i \leftrightarrow r_j)$$

- Unimodular

$$P^{-1}(\gamma r_i) = P\left(\gamma^{-1} r_i\right)$$

$$P^{-1}((\varphi)r_i + r_j) = P((-\varphi)r_i + r_j)$$

- Transformation

$$P(r_i \leftrightarrow r_j) = P(c_i \leftrightarrow c_j), \quad P(\gamma r_i) = P(\gamma c_i),$$

However, $P((\varphi)r_i + r_j) = P((\varphi)c_j + r_i)$.

2.6 Unimodular matrix polynomials

Z. Song

- Elementary operations

Theorem 2.21 Performing an elementary row operation on an $m \times n$ λ-matrix $A(\lambda)$ is equivalent to premultiplying $A(\lambda)$ by the corresponding elementary matrix in $P(r_i \leftrightarrow r_j)$, $P(\gamma r_i)$ and $P((\varphi)r_i + r_j)$. Similarly, Performing an elementary column operation on an $m \times n$ λ-matrix $A(\lambda)$ is equivalent to postmultiplying $A(\lambda)$ by the corresponding elementary matrix in $P(c_i \leftrightarrow c_j)$, $P(\gamma c_i)$ and $P((\varphi)c_i + c_j)$.

2.6 Unimodular matrix polynomials

Z. Song

- Equivalent matrix polynomials

Definition 2.15 $A(\lambda)$ is said to be equivalent to $B(\lambda)$ if $B(\lambda)$ is obtained by performing a finite series row or/and column operations on $A(\lambda)$, denoted $A(\lambda) \cong B(\lambda)$.

Theorem 2.22 $A(\lambda)$ and $B(\lambda)$ are equivalent if and only if there exist unimodular λ-matrices $P(\lambda)$ and $Q(\lambda)$, such that

$$B(\lambda) = P(\lambda)A(\lambda)Q(\lambda)$$

2.6 Unimodular matrix polynomials

Z. Song

- Any matrix polynomials can be transformed into the Smith cannonical form via the a finite sequence of row and column operations.

- Before we give the Smith cannonical from, we first give a lemma which is useful in transforming matrix polynomials into the Smith cannonical from.

Lemma 2.1 Suppose that the entry $a_{11}(\lambda)$ in the upper-left corner of a λ-matrix $A(\lambda)$ is nonzero and there is at least one entry in $A(\lambda)$ that is not divisible by $a_{11}(\lambda)$, then there exists a λ-matrix $B(\lambda)$ equivalent to $A(\lambda)$ whose upper left corner is nonzero but with a degree less than the degree of $a_{11}(\lambda)$.

2.6 Unimodular matrix polynomials

Z. Song

- Smith cannonical form (pseudo-diagonal polynomial matrix)

Theorem 2.23 Let $A(\lambda) \in \mathcal{C}^{m\times n}[\lambda]$, rank$A(\lambda) = r$, then

$$A(\lambda) \cong \begin{bmatrix} D(\lambda) & 0 \\ 0 & 0 \end{bmatrix} \qquad (1)$$

$$D(\lambda) = \text{diag}\{\, d_1(\lambda)\,, d_2(\lambda)\,, \ldots\,, d_r(\lambda)\,\} \qquad (2)$$

where $d_i(\lambda)$ is monic and satisfies

$$d_i(\lambda)\,|\,d_{i+1}(\lambda)\,, \quad i = 1, 2, \ldots, r-1\,.$$

- Proof:Pg 69

Definition 2.16 The equivalent form (1) is referred to as the Smith canonical form of $A(\lambda)$, $d_1(\lambda)$, $d_2(\lambda)$, ..., $d_r(\lambda)$ are called the invariant polynomials of $A(\lambda)$.

- Example 2.16 on Pg 70.

2.6 Unimodular matrix polynomials

Z. Song

- Smith cannonical form is unique, to show it, the following concept of k-th order divisors is needed.

Definition 2.17 Let the λ-matrix $A(\lambda)$ be of rank r. For k, $1 \leq k \leq r$, $A(\lambda)$ has at least one nonzero minor of order k. The divisor of kth order minors is defined as the greatest common divisor of all nonzero kth order minors of $A(\lambda)$.

Theorem 2. 24 Equivalent λ-matrices have the same rank and the same divisor of kth order minors.

Theorem 2. 25 The Smith canonical form of a λ-matrix $A(\lambda)$ is unique.

2.6 Unimodular matrix polynomials

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- Smith cannonical form is unique

Theorem 2.25 The Smith canonical form of a λ-matrix $A(\lambda)$ is unique.

Let the Smith canonical form of a λ-matrix $A(\lambda)$ be

$$\begin{bmatrix} d_1(\lambda) & & & & & & \\ & d_2(\lambda) & & & & & \\ & & \ddots & & & & \\ & & & d_r(\lambda) & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{bmatrix}$$

where $d_1(\lambda), d_2(\lambda), \ldots, d_r(\lambda)$ are all monic and $d_i(\lambda)|d_{i+1}(\lambda)$. It is clear that the divisor of the kth order minors is

$$D_k(\lambda) = d_1(\lambda)d_2(\lambda)\cdots d_k(\lambda)\,, \qquad (1)$$

Then please see the proof on Pg 74.

- Proof:Pg 74

2.6 Unimodular matrix polynomials

Z. Song

- Applying the uniqueness of the Smith canonical form, we obtain
  - Necessary and sufficient conditions for equivalent matrix polynomials

Theorem 2.26 λ-matrices $A(\lambda)$ and $B(\lambda) \in \mathcal{F}^{m\times n}$ are equivalent, if and only if for any $k > 0$, $A(\lambda)$ and $B(\lambda)$ have the same divisor of the kth order minors.

Theorem 2.27 λ-matrices $A(\lambda)$ and $B(\lambda) \in \mathcal{F}^{m\times n}$ are equivalent, if and only if they have the same invariant polynomials.

2.6 Unimodular matrix polynomials

Z. Song

- Applying the uniqueness of the Smith canonical form, we obtain

  - Necessary and sufficient conditions for invertible matrix polynomials (unimodular matrices)

Corollary 2.6 A λ-matrix $A(\lambda)$ is invertible if and only if it is equivalent to an identity matrix.

Corollary 2.7 A λ-matrix $A(\lambda)$ is invertible if and only if $A(\lambda)$ can be represented as the product of a finite sequence of elementary matrices.

2.7 Elementary divisors and equivalence of matrix polynomials

Z. Song

- Definition of elementary divisors

Let $A(\lambda) \in \mathcal{C}^{m\times n}[\lambda]$, $\mathrm{rank}A(\lambda) = r$, then

$$A(\lambda) \cong \begin{bmatrix} D(\lambda) & 0 \\ 0 & 0 \end{bmatrix} \tag{1}$$

$$D(\lambda) = \mathrm{diag}\{\, d_1(\lambda)\,, d_2(\lambda)\,, \ldots\,, d_r(\lambda)\,\} \tag{2}$$

where $d_i(\lambda)$ is monic and satisfies

$$d_i(\lambda)\,|\,d_{i+1}(\lambda)\,, \quad i = 1, 2, \ldots, r-1\,.$$

the invariant polynomials $d_k(\lambda)$ above can be decomposed as

$$\begin{aligned} d_1(\lambda) &= (\lambda-\lambda_1)^{k_{11}}(\lambda-\lambda_2)^{k_{12}}\cdots(\lambda-\lambda_l)^{k_{1l}} \\ d_2(\lambda) &= (\lambda-\lambda_1)^{k_{21}}(\lambda-\lambda_2)^{k_{22}}\cdots(\lambda-\lambda_l)^{k_{2l}} \\ &\vdots \\ d_r(\lambda) &= (\lambda-\lambda_1)^{k_{r1}}(\lambda-\lambda_2)^{k_{r2}}\cdots(\lambda-\lambda_l)^{k_{rl}} \end{aligned}$$

2.7 Elementary divisors and equivalence of matrix polynomials

- Definition of elementary divisors-continued

Since $d_k(\lambda)|d_{k+1}(\lambda)\ \forall k = 1, 2, \ldots, r$, we must have

$$k_{1j} \leq k_{2j} \leq \cdots \leq k_{rj}\,,\ \forall k = 1, 2, \ldots, r$$

here λ_1, λ_2, ..., λ_l are the distinct roots of $d_r(\lambda)$, and k_{r1}, k_{r2}, ..., k_{rl} are all positive integers. However, some of k_{1j}, k_{2j}, ..., $k_{r-1,j}$ may be zeros. For a fixed j, if $k_{ij} = 0$, we must have $k_{1j} = 0$, $k_{2j} = 0$, ..., $k_{i-1,j} = 0$. Referring to the following factors

$$\begin{cases} (\lambda-\lambda_1)^{k_{11}}\,, & (\lambda-\lambda_2)^{k_{12}}\,, & \cdots\,, & (\lambda-\lambda_l)^{k_{1l}} \\ (\lambda-\lambda_1)^{k_{21}}\,, & (\lambda-\lambda_2)^{k_{22}}\,, & \cdots\,, & (\lambda-\lambda_l)^{k_{2l}} \\ \vdots & & & \\ (\lambda-\lambda_1)^{k_{r1}}\,, & (\lambda-\lambda_2)^{k_{r2}}\,, & \cdots\,, & (\lambda-\lambda_l)^{k_{rl}} \end{cases} \tag{1}$$

we call the totality of the factors $(\lambda-\lambda_j)^{k_{ij}}$ with $k_{ij} > 0$ the elementary divisors of $A(\lambda)$.

2.7 Elementary divisors and equivalence of matrix polynomials

Z. Song

- Remark

  - Equivalent matrix polynomials have the same invariant polynomials, hence the same elementary divisors. (necessary condition)

  - Two matrix polynomials with the same elementary divisors may not be equivalent (not sufficient condition)

    - Reason:

    - 1. elementary divisors do not count constant

    - 2. different invariant polynomials may have the same elementary divisors

- Please refer to the example on Pg 75.

2.7 Elementary divisors and equivalence of matrix polynomials

Z. Song

- Properties of elementary divisors

Theorem 2.29 The totality of the elementary divisors of the block diagonal λ-matrix $A(\lambda)$ with the block diagonals $B(\lambda)$ and $C(\lambda)$

$$A(\lambda) = \begin{bmatrix} B(\lambda) & 0 \\ 0 & C(\lambda) \end{bmatrix}$$

is the totality of all the elementary divisors of $B(\lambda)$ and $C(\lambda)$.

- Please read the proof after class, let's have a look at Example 2.24 for illustrating this theorem statement.

- This theorem can be generalized to a block diagonal matrix polynomial with any finite number of block diagonals.

2.7 Elementary divisors and equivalence of matrix polynomials

Z. Song

- Properties of elementary divisors-continued

Theorem 2.30 The totality of all the elementary divisors of the λ-matrix

$$A(\lambda) = \begin{bmatrix} B_1(\lambda) & & & \\ & B_2(\lambda) & & \\ & & \ddots & \\ & & & B_t(\lambda) \end{bmatrix}$$

is the totality of all the elementary divisors of $B_1(\lambda)$, $B_2(\lambda)$, ..., $B_t(\lambda)$.

Theorem 2.31 The totality of the elementary divisors of the λ-matrix

$$A(\lambda) = \begin{bmatrix} f_1(\lambda) & & & & & \\ & f_2(\lambda) & & & & \\ & & \ddots & & & \\ & & & f_r(\lambda) & & \\ & & & & 0 & \\ & & & & & \ddots \\ & & & & & & 0 \end{bmatrix}$$

is the totality of all the powers of polynomials of degree 1 contained in $f_1(\lambda)$, $f_2(\lambda)$, ..., $f_r(\lambda)$.

- A special case

2.7 Elementary divisors and equivalence of matrix polynomials

- Example 2.25

  - A square matrix A
  - Eigenmatrix of A

  Characteristic matrix $A(\lambda) = \lambda E - A$ is a special kind of λ-matrices.